

EFA-GUI User Manual

The objective of EFA-GUI is to transition from manual single-particle analysis to an automated, high-throughput workflow. Users can perform statistical shape analysis, harmonic sensitivity studies, and contour reconstruction using an interactive Python-based Streamlit graphical user interface (GUI).

The software is available through two deployment modes:

- **Web Application:** <https://efa-gui.streamlit.app/>
- **Local Installation (PyPI):** <https://pypi.org/project/efa-gui/>

Both deployment modes provide identical functionality.

Option A: Use the Web Application

1. Open the application:
 - <https://efa-gui.streamlit.app/>
 - If the application is inactive, wait for it to wake up.
2. Upload your prepared particle dataset and specify the image scaling factor (px/mm).
3. Alternatively, download and use the example dataset:
 - test_data.csv
 - GitHub repository: <https://github.com/aidanaboribayeva/Computational-Elliptic-Fourier-Analysis>
4. For the provided test dataset, use:
 - Scaling factor: **77 px/mm**

Option B: Install and Run Locally

Quick Installation

Open a terminal or command prompt and run:

```
pip install efa-gui
```

Verify the installation:

```
efa-gui-self-test
```

Launch the application:

```
efa-gui
```

Recommended Installation Using a Virtual Environment (PowerShell)

```
python -m venv env  
Set-ExecutionPolicy -Scope Process -ExecutionPolicy Bypass
```

```
.\env\Scripts\Activate.ps1
python -m pip install --upgrade pip
python -m pip install efa-gui
python -c "import efa_gui; print(efa_gui.__version__)"
efa-gui-self-test
efa-gui
```

Installation in Jupyter or Notebook Environments

```
python -m pip install efa-gui
python -c "import efa_gui; print(efa_gui.__version__)"
efa-gui-self-test
efa-gui
```

Automated Self-Test

To verify that the software is installed correctly and functioning as expected, run:

```
efa-gui-self-test
```

The automated self-test loads the sample dataset, executes Module 3, and compares the generated shape indices against validated reference values.

The test is successful if the following message is displayed:

```
Self-test passed: sample CSV loaded, Module 3 ran, and reference indices matched.
```

Workflow

Module 1: Reconstruction

- Select an individual particle.
- Adjust the harmonic order slider from 1 to 40.
- Observe the reconstruction error and how the reconstructed contour converges toward the original particle shape.

Module 2: Sensitivity Analysis

- Apply different harmonic orders to the entire dataset.
- Evaluate how shape descriptors, such as Asymmetry, stabilize as the number of harmonics increases.

Module 3: Statistical Analysis

- Generate distribution plots for:
 - Elongation
 - Angularity
 - Roughness
- Export the correlation matrix to identify potentially redundant shape descriptors.

Important Notes

Streamlit Wake-Up

After periods of inactivity, the hosted Streamlit application may enter a sleep state. When visiting the web application, you may need to click:

"Yes, get this app back!"

and wait a few moments for the application to restart.

Input Data Requirements

The application accepts contour datasets in **.csv**, **.txt**, **.tsv**, **.xlsx**, and **.xls** formats. Columns may be separated by **commas**, **tabs**, or **whitespace**, and the software automatically detects the delimiter during import.

Each dataset must contain the following columns:

- **ID** – unique particle (object) identifier
- **X** – x-coordinate of the contour point
- **Y** – y-coordinate of the contour point

The contour coordinates for each particle **must be provided in sequential order and represent a complete closed contour**. Since Elliptic Fourier Analysis (EFA) is mathematically defined for closed boundaries, incomplete or open contours may lead to incorrect Fourier descriptors and unreliable shape indices. If contour coordinates are obtained from image-processing software, ensure that the exported outline forms a closed boundary before importing the dataset into the application.

Processing Time

For larger datasets (approximately 100 particles or more), increasing the harmonic order may increase computation time. Wait until the **"Running"** indicator in the upper-right corner disappears before proceeding with further analyses.

Citation

If you use this software in your research, please cite:

Boribayeva, A., Sultaniyar, S., Lukmanov, I., Baigarina, A., Rojas-Solórzano, L. R., Curtis, J. S., Govender, N., & Golman, B. (2026). *Integrated characterization, classification, and quasi-3D reconstruction of highly irregular particles using multiscale shape descriptors for predictive DEM flow simulation*. Powder Technology, 122435. DOI: 10.1016/j.powtec.2026.122435